

Christopher Felix Atika Portfolio

Christopher Felix Atika

iOS & Android Engineer @  traveloka

Portfolio Website

Link

<https://christopheratika.site>

Github

Link

<https://github.com/lixkelik>

Major

Computer Science @ Binus University

Competition

Infarm

CONCEPT

Project 1 of 5

Farmers in Indonesia faced some challenges regarding farm input prices because of COVID-19. Farm input prices are easily manipulated as Indonesian farmers don't have many connections to other sources of farm inputs. They don't have the option to choose a different vendor. Infarm comes to fix those issues by building an e-commerce site designed for farmers.

Year
2023

Infarm was created with consideration for its user-friendliness, as we will predict that many farmers are not fluent with technology. Also, we created a subcategory product for them to easily find what their needs are based on our subcategory features.

Infarm also has message with seller features, which help farmers negotiate and talk to the seller directly. Infarm is built using Flutter as the framework and Firebase as the database service. We also use VelocityX for our state management.

Category
Competition

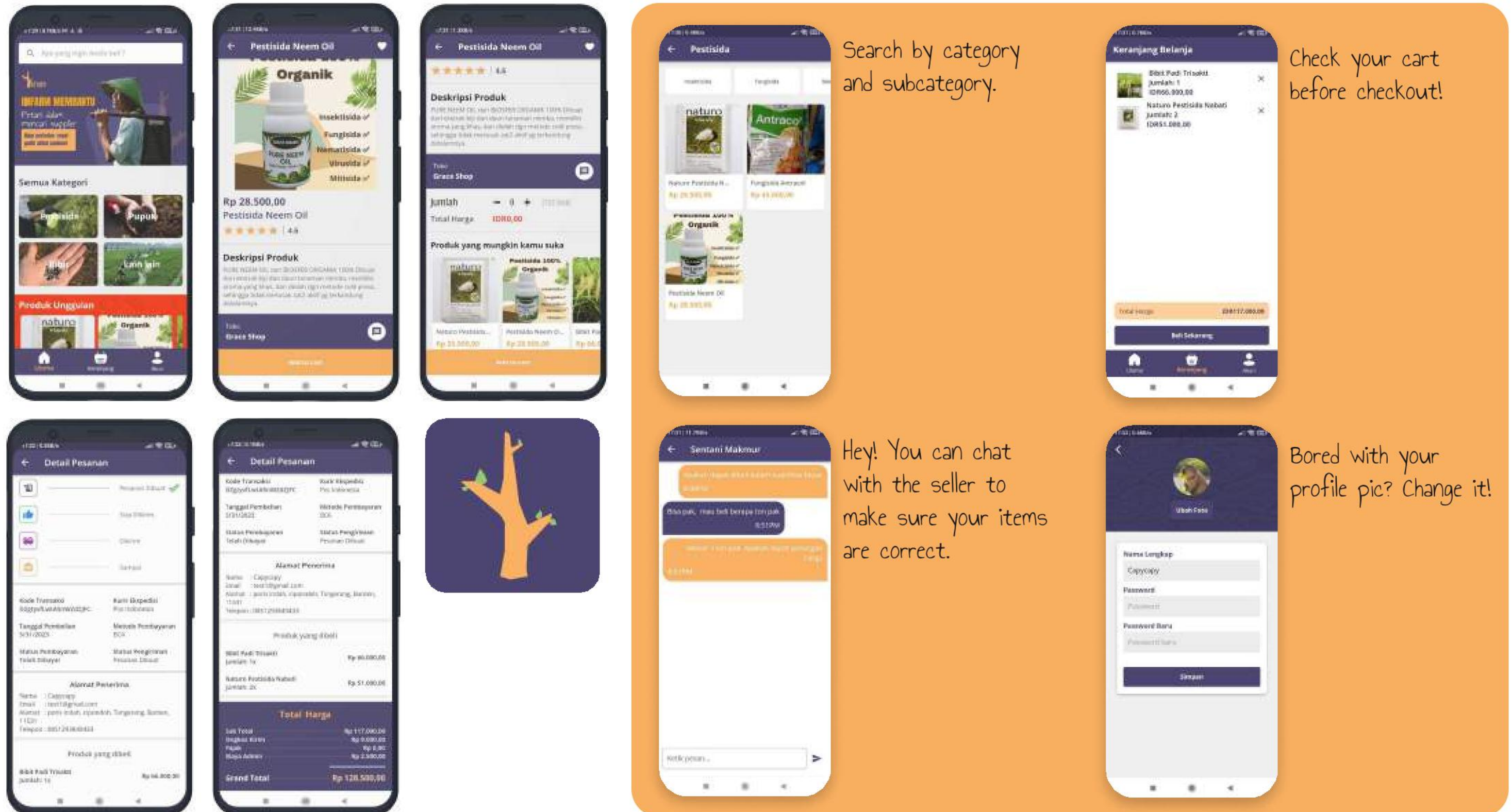
Team

Application Link

[Apk File](#)

Source code

<https://github.com/lixxelik/infarm>



Check your cart before checkout!

Hey! You can chat with the seller to make sure your items are correct.

Bored with your profile pic? Change it!

Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

[in](#) Christopher Felix Atika

<https://christopheratika.site>

Seekers

CONCEPT

Project 2 of 5

Year
2023

Category
Competition

Team

Application Link

[Apk File](#)

[Proposal Link](#)

Source code

<https://github.com/lixkelik/seekers>

How do blind people, especially blind children, learn an object at such a young age? This question led us to a problem and brought out my and my partner's curiosity. They are missing the context of an object or the complete meaning of an object, which we call lexical form. Inclusive problems arise too, as they can't completely communicate with their friends because of their social skills or play games collaboratively. This problem could possibly be fixed by a software approach by creating an inclusive education with gamification.

The software approach that we use is to create an application that can help blind children learn, be able to know, describe, and collaboratively play with their friends, and understand what their friends think about an object they scan by using object detection. Yes, we implemented an object detection AI in the Seekers Android application. We used free models available at the online Tensorflow library.

Seekers was built using Flutter and Firebase as its databases. The plugin used is Tensorflow Lite, which helped us implement Tensorflow in the application. Seekers have several features, such as an exp system, badges, object detection, and collaborative games. This collaborative game will push blind kids to touch the object and require their friends to play by giving them the game code, and their friends can give their thoughts about the object.

Seekers has two roles:
visually impaired kids and
sighted peers.



Visually impaired kids can hear
their friend's description of
the object that they scanned.

Their friends can
input their knowledge
about the object.

We have built-in
speech to text to
help them write
without using
keyboard

After the visually impaired kids have
scanned five objects, they can give the
code to their friend.



Seekers

An educational app for
visually impaired children to
create inclusion at their
younger age



Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

 [Christopher Felix Atika](#)

<https://christopheratika.site>

Year
2023

Category
Competition

Team

Did you know that in 2019, there were 150 thousand hectares of destroyed and displaced agricultural land? Agricultural land that switches roles every year, equivalent to 300,000 tons of products presently! Yes, Indonesia's agricultural land is shrinking year after year. All this is due to insufficient capital, which leads to the effectiveness of agricultural processes. As the efficiency decreases, the profits also decrease, so that the entrepreneur is more interested in other more profitable business models.

Indonesia is an agricultural country. This means that the economy of Indonesia is largely supported by agricultural or agricultural activities. Indonesia has a fertile soil and also a tropical climate that causes sufficient sunlight and high rainfall intensity. According to data from the World Bank, between 2010 and 2016, the agricultural sector alone contributed 13.86% of the national GDP. Agricultural outcomes contribute to GDP in the form of food availability, foreign currency income, employment, and raw material production. Along with the large contribution of agriculture to GDP, the allocation of government funds to this sector is also large, with an average of 31.24% of the state budget. This shows that the government also supports the sustainability of the agricultural sector. However, Indonesia is an increasingly threatened agricultural country. This is due to the uncontrolled shrinkage of farmland.



Taninvest
Crowdfunding with third party mentor and supervisor for Indonesian farmers!
Aim to increase farm land.



Visio

CONCEPT

Project 4 of 5

Visio ideas are just the same as Seekers, but Visio is more enhanced, user-friendly, and eye-catching than Seekers. Visio also have added features such as learn Braille so children can learn braille using haptic feedback, and a chatbot. This app now aims not only at visually impaired children but also at children who are curious and want to learn anything. With ChatGPT integration, they can ask anything related to objects, so they can have more knowledge to help visually impaired children.

Visio uses isolate or multithreading in Java, so it will create a more smooth user experience for them to use our application while the AI processes and detects the object that comes from the camera.

Year

2023

Category
Competition

Team

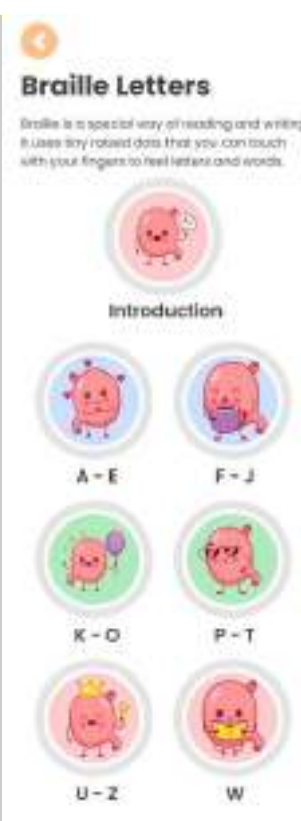
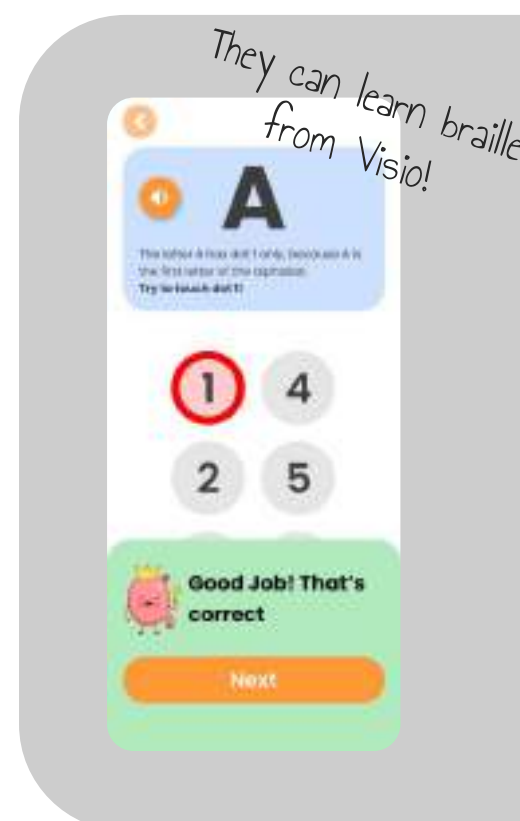
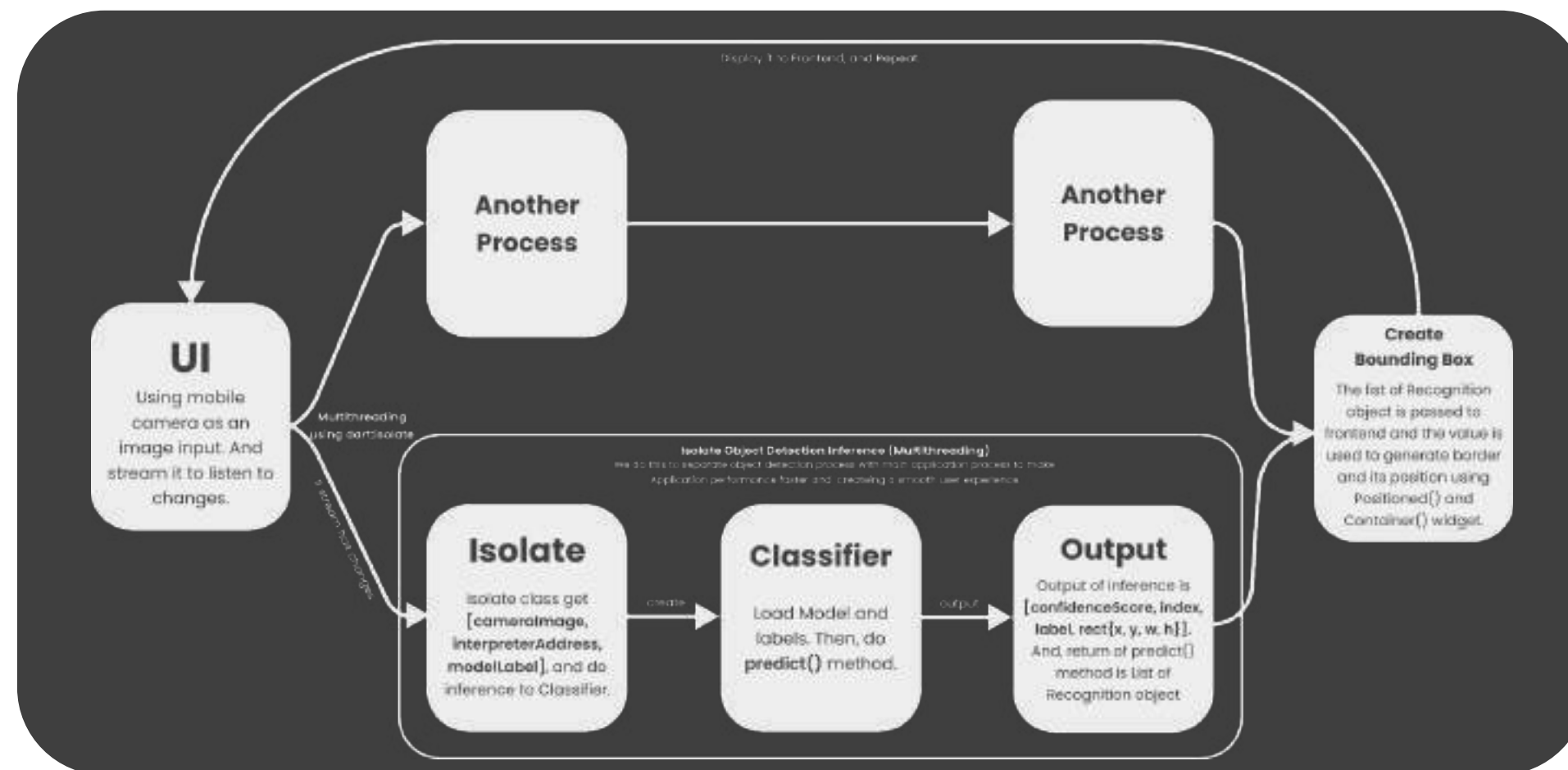
Application Link

Apk File

Proposal Link

Source code

<https://github.com/lixkelik/Visio>



ChatGPT Integration.
So children can ask whatever they wanted to know about something or an object.



Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

[Christopher Felix Atika](#)

<https://christopheratika.site>

Aura

CONCEPT

Project 5 of 5

Aura is an application for the visually impaired to find suitable and matching outfits, including tops, bottoms, and outerwear, based on the purpose or activity they are going to engage in. Using AI technology, Aura will make suggestions by describing items and finding appropriate outfit combinations depending on the kind, color, and pattern of the apparel. Furthermore, Aura will make use of RFID technology to help blind people find and get clothes out of closets full of similar but different items.

Year

2024

Category
Competition

Team

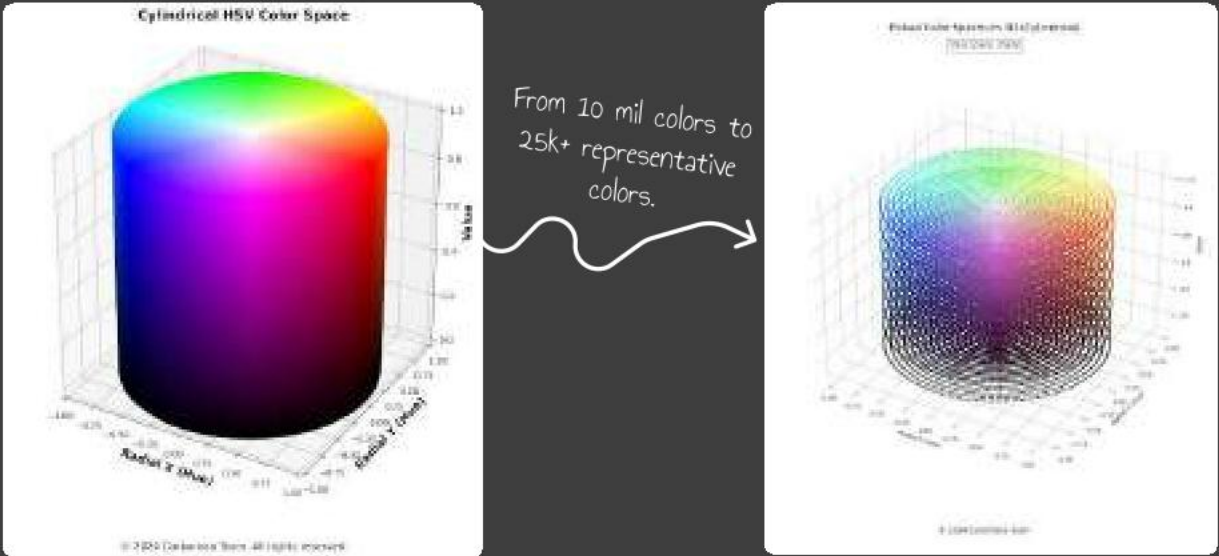
Supporting Link

Coming soon to Playstore
and Appstore

[Aura Paper](#)

What is Harmony AI, the backbone of Aura?

Harmony AI is our way to determine whether a combination of two or more colors matches or not. It also provides users with information on whether the color combination gives a neutral, vibrant, or dark accent vibe. We developed our own algorithm to generate the full color spectrum visible to the human eye by mapping colors into a readable and easy-to-calculate format using a cylindrical HSV model. Then, we traverse through the entire circle in 2-degree increments, including light and dark variations, to capture all points that represent each color. This approach generates a dataset of over 175,000 colors.



Then..

We generate pairs of matching colors for each representative color point using principles from color theory, aiming to achieve what is known as color harmony. Color harmony is the ideal state where a combination of colors is visually balanced and pleasing. To reach this harmony, we apply three rules from color theory: Complementary, Analogous, and Monotone Chromatic.



The final dataset generated using this algorithm contains approximately **175,172** entries.

TensorFlow

Flutter



Aura won 3rd place at Gemastik XVII 2024.



Aura won 1st place at Microsoft AI4A APAC Hackathon 2024

Judged by CFO Microsoft Indonesia and Founder of Suarise.



Presenting Aura at Microsoft HQ Indonesia. Attended by Director, CFO and all Employee of Microsoft Indonesia.

Scan to learn more



linktr.ee/project_aura



Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

[in](#) Christopher Felix Atika

<https://christopheratika.site>

Course Project

Manu

CONCEPT

Project 1 of 3

Indonesia's UMKM, known as Small and Medium Enterprise (SME), is rising exponentially after COVID-19. But, even though it is rising, the chance for its UMKM to survive is at a low percentage. From the data we gather, the main reason they cannot survive is their lack of capital to build their businesses and survive. Thus, Manu proposed that to solve it, we help UMKM in Indonesia search for available sharing factories based on their needs.

Year

2023

Manu is a one-of-a-kind idea that is new and has never been proposed before. Manu will work by creating a factory or getting a partnership with another factory to produce something based on UMKM's needs or preferences. Manu has a mobile application to make it easier for the UMKM to connect and do transactions with the factory.

With the Manu mobile application, they can search a factory based on the category and place as we create a filter feature. With Manu, we hope that they can continue to grow their UMKM significantly with less capital.

Category

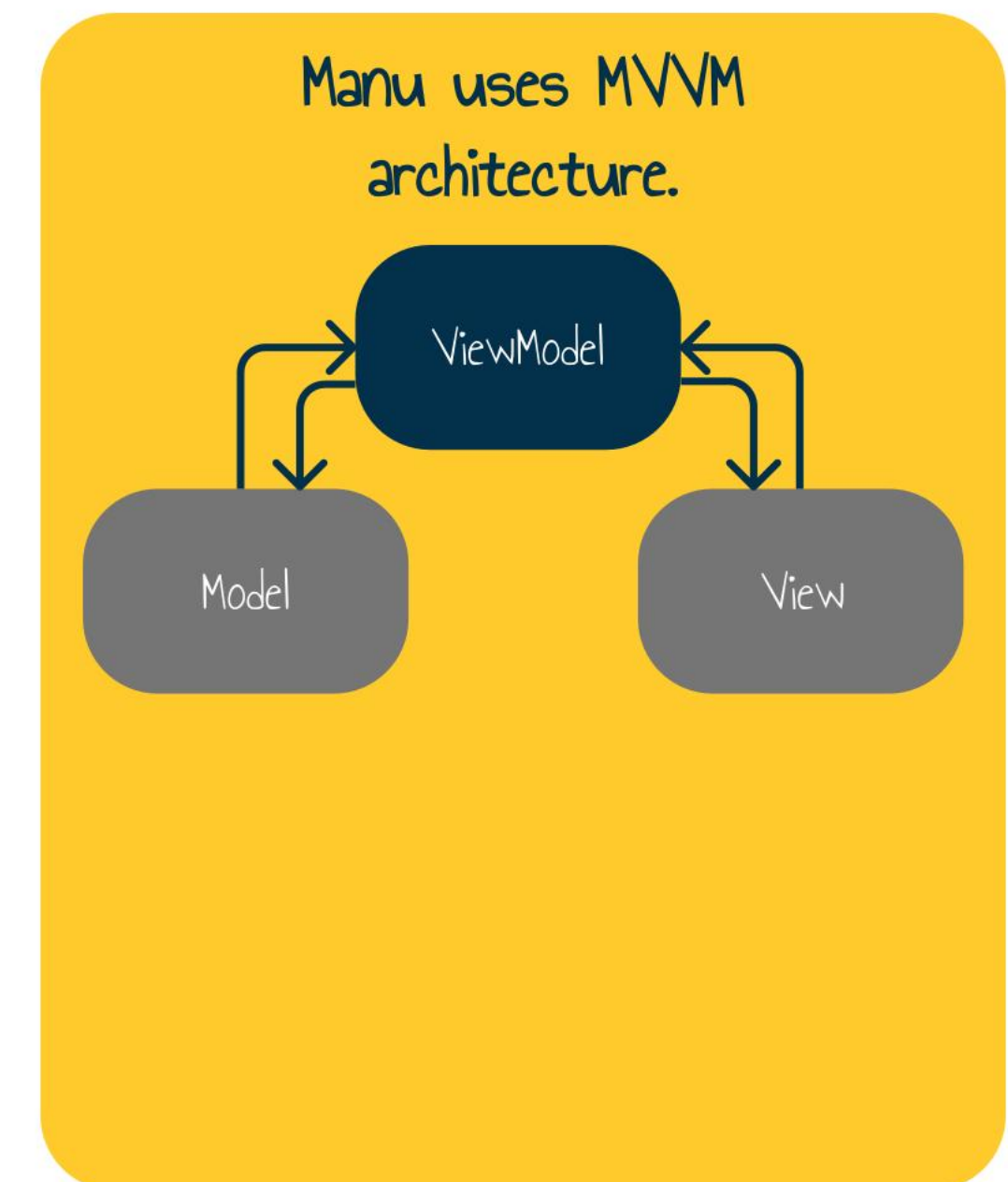
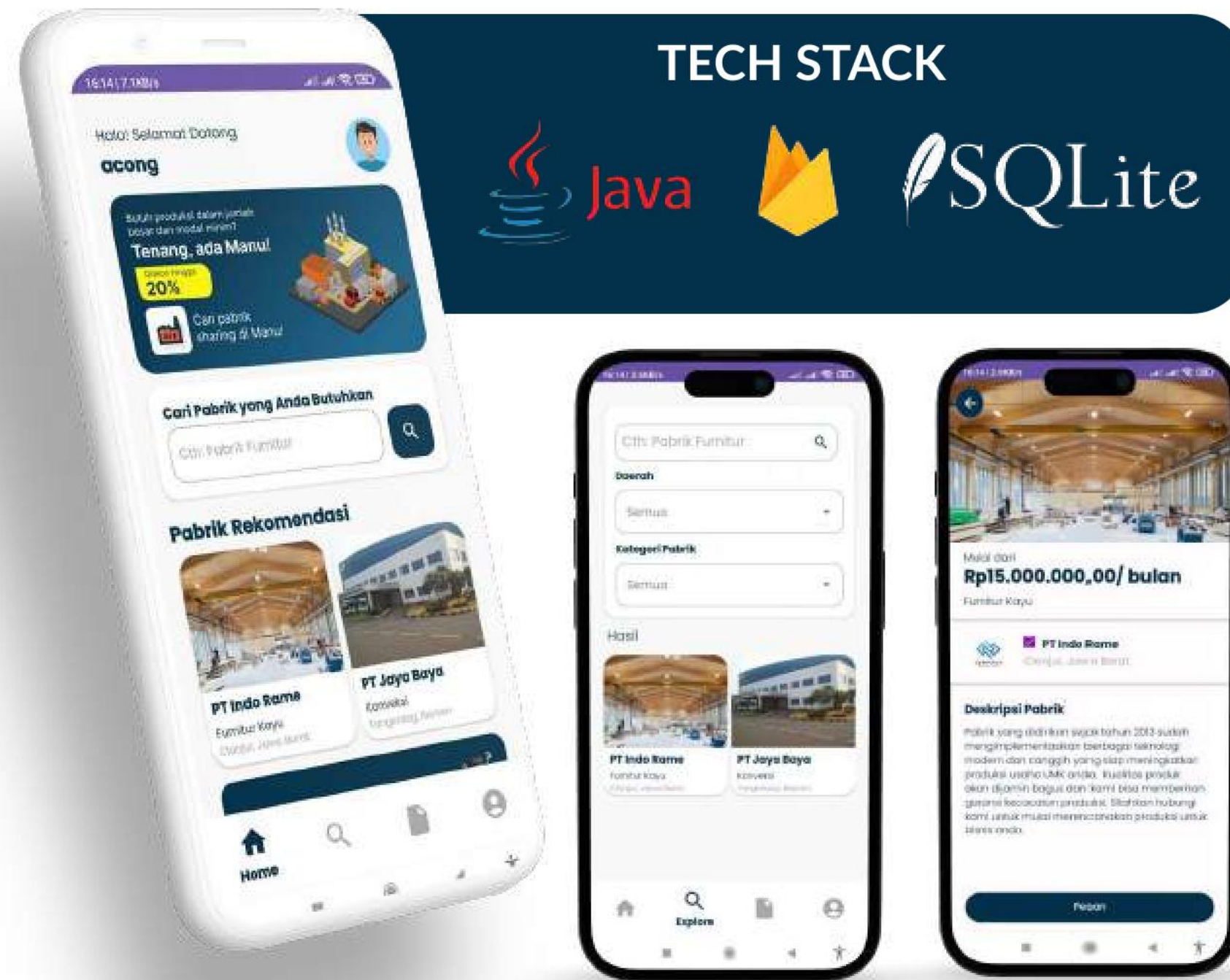
Course Project

Team

Application Link

Source code

<https://github.com/lixxelik/Manu>



Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

[in](https://www.linkedin.com/in/christopherfelixatika/) Christopher Felix Atika

<https://christopheratika.site>

Tukerin

CONCEPT

Project 2 of 3

The transition from conventional vehicles to electric ones requires a lot of challenge and consideration from the customer itself. Indonesian citizens always consider the charging time and the infrastructure's availability. But we found a solution for that by introducing Tukerin, a mobile application to find battery swap stations (BSS) and swap them instantly from the application without waiting a few hours.

Year

2023

What makes Tukerin different from their competitor is that Tukerin uses a customer-to-customer business model. So, normal EV users can create a BSS and register it with our application. This will help BSS grow rapidly without the big capital investment needed to create a BSS. We think of it just like an online food app where small restaurants can register their restaurant for online food delivery.

With Tukerin, we have a vision to make motorcycle EVs grow and remove any doubts from the customer perspective regarding charging time and the infrastructure, where they still have stigmas like "Indonesia is not ready for EV transformation".

Category

Course Project

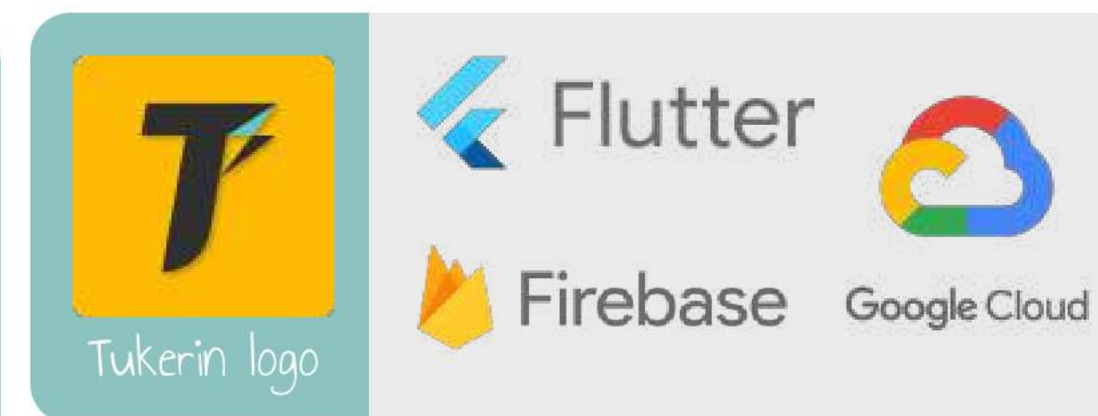
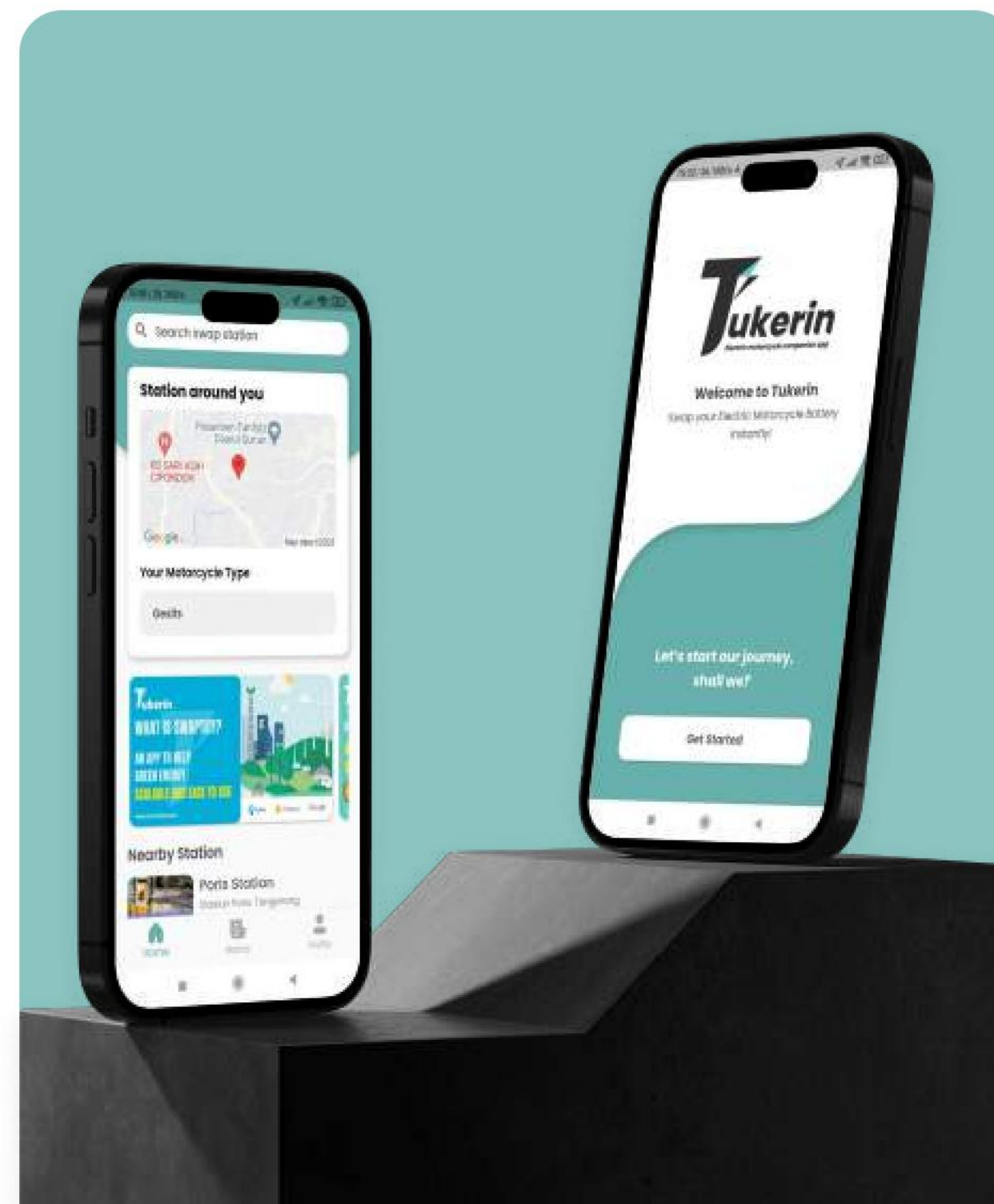
Individual

Application Link

[Apk File](#)

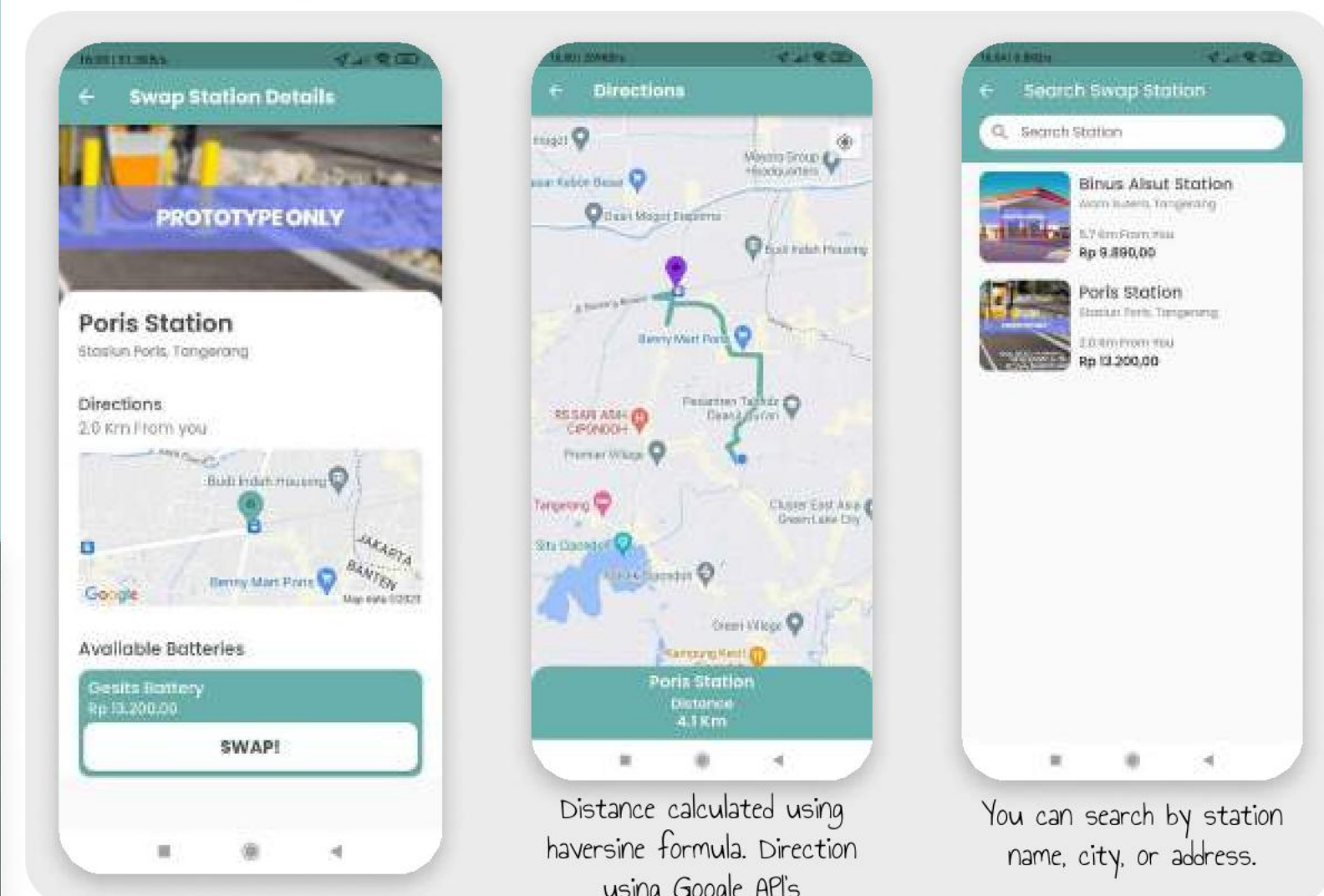
Source code

<https://github.com/lixkelik/tukerin>



Indexed at IEEE Xplore

[Click Here to see the publication](#)



Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

[in](#) Christopher Felix Atika

<https://christopheratika.site>

Adex Legends

CONCEPT

Project 3 of 3

Adex Legends is a simple front-end website to fulfill my Human-Computer Interaction courses. This simple website is built for a top game webpage so that users can see what the games are about and a list of the Legends available in Adex Legends.

Adex Legends is built with CSS and HTML, and the language used is Javascript. There is no functionality with this website as it is only for the frontend and to learn how to create a form text field with HTML and some simple validation in it.

Year
2022

Category
Course Project

Individual

Source Code

<https://github.com/lixkelik/Adex-Legends>

[Figma Link](#)



Christopher Felix Atika

christopherfelix.atika@gmail.com

+62 851 5510 0892

 [Christopher Felix Atika](#)

<https://christopheratika.site>

External Project

The CRR Management App

Year
2023

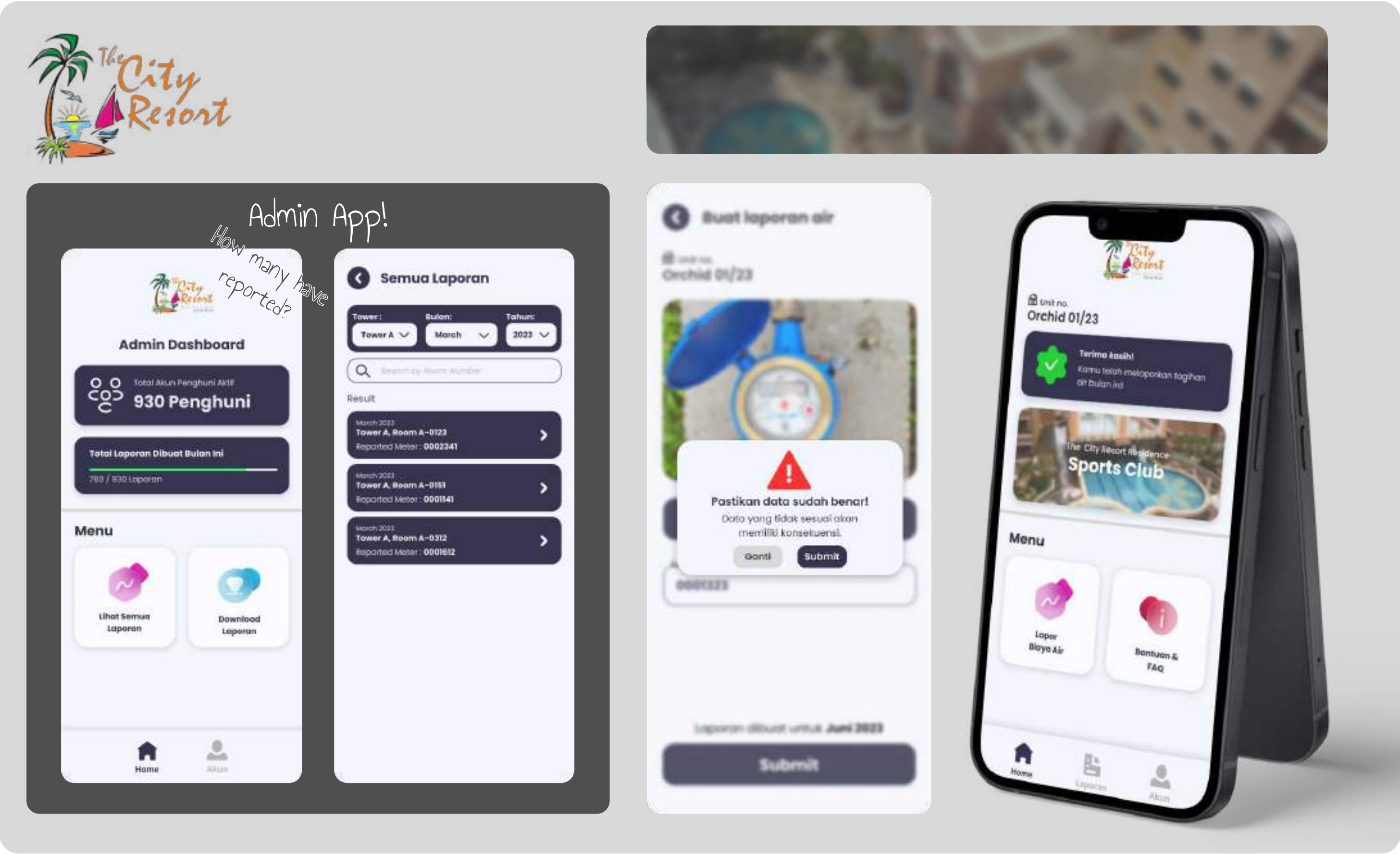
Category
External Project

Individual

CONCEPT

The City Resort Residence is an apartment in West Jakarta. This Android application will focus on solving their problem with monitoring and gathering the report for the water meter each month. This mobile app can hopefully reduce dishonest reports from the tenant by implementing a camera on the mobile phone rather than manual reporting.

The CRR Mobile App will gradually release new features if asked for by stakeholders. It also has two roles, one for the tenant and the other for the admin to download the report in csv files and monitor the tenant's reporting activity.



Christopher Felix Atika

christopherfelix.atika@gmail.com
+62 851 5510 0892
 Christopher Felix Atika
<https://christopheratika.site>